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(54) **POROUS POLYMER SEPARATOR LAYER HAVING A NON-UNIFORM CROSS-SECTIONAL THICKNESS FOR USE IN A SECONDARY LIQUID-ELECTROLYTE BATTERY**

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H01M 2/18 (2006.01)

(52) **U.S. Cl.**
USPC **429/246**; 429/247; 429/254

(58) **Field of Classification Search**
None
See application file for complete search history.

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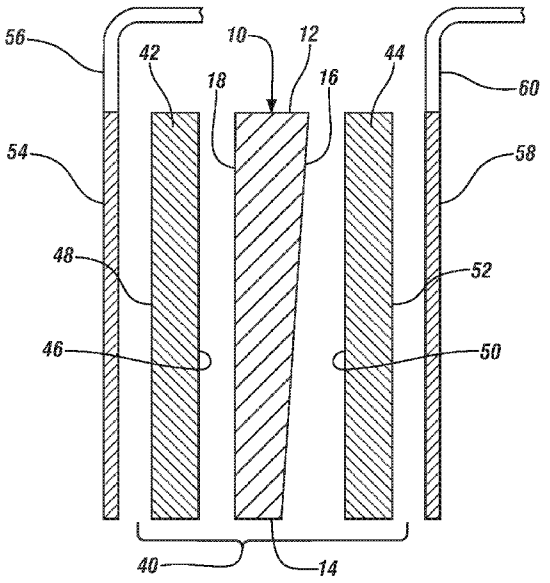
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(57) **ABSTRACT**

A porous polymer separator layer that exhibits a non-uniform cross-sectional thickness and a method of making the same are disclosed. The porous polymer separator layer may be made by a process that involves forming a film having a non-uniform cross-sectional thickness similar to that sought to be imparted to the resultant separator layer and deriving the porous polymer separator layer from the film. An electrochemical battery cell for a secondary liquid-electrolyte battery may incorporate the disclosed porous polymer separator layer between a negative electrode and a positive electrode in a way that helps maintain a more evenly distributed current density within the cell.

20 Claims, 4 Drawing Sheets



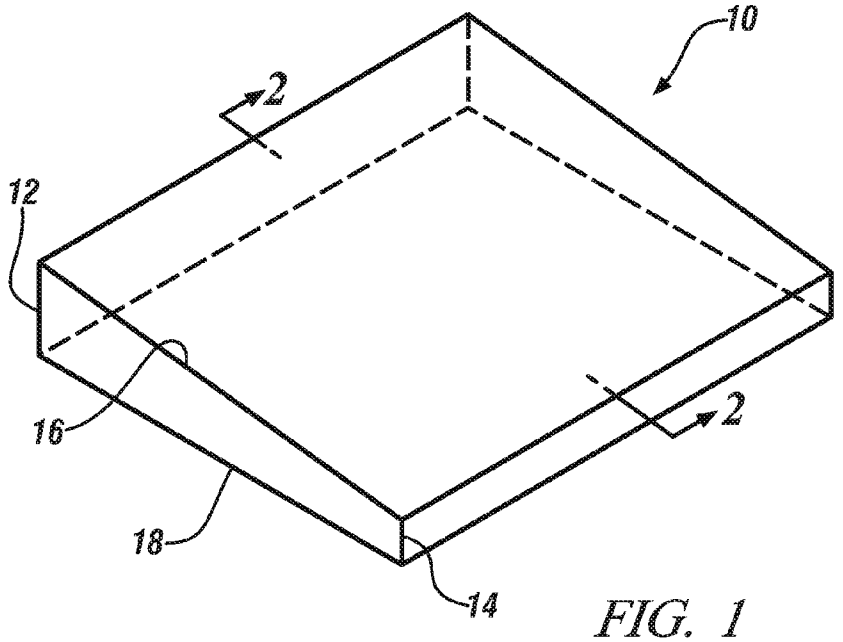


FIG. 1

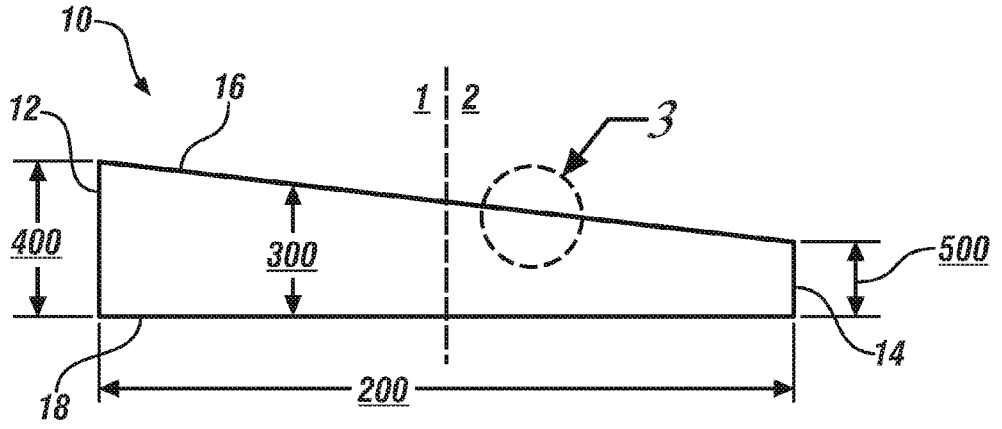


FIG. 2

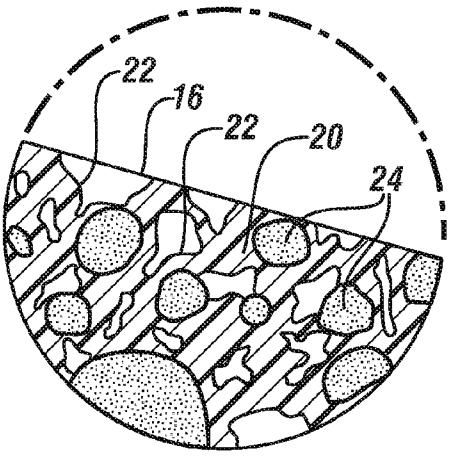


FIG. 3

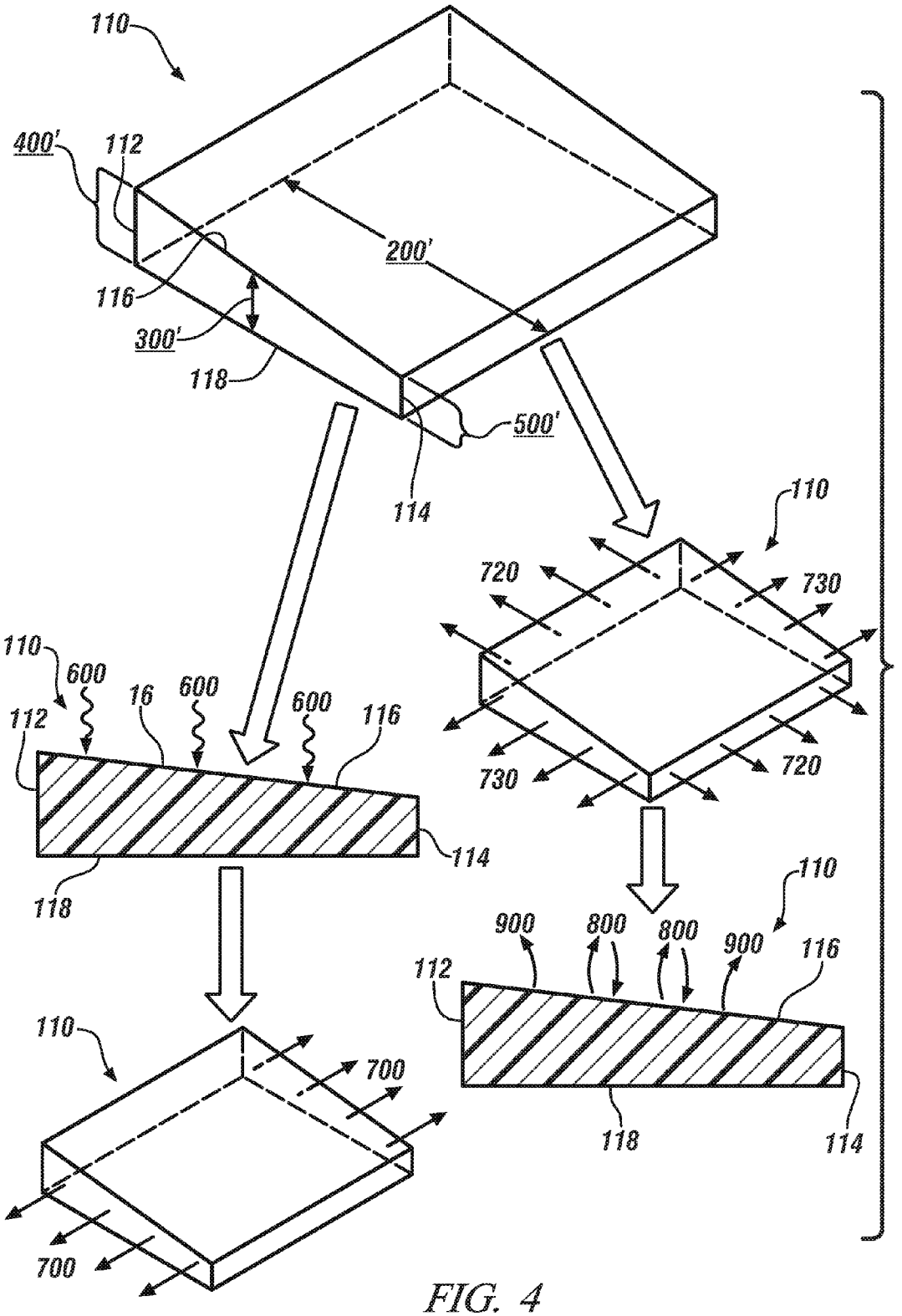

110
400'
112
200'
116
300'
118
500'
114
110
720
730
730
720
110
600
16
600
600
116
112
114
118
110
900
800
800
900
116
112
118
114
110
700
700

FIG. 4

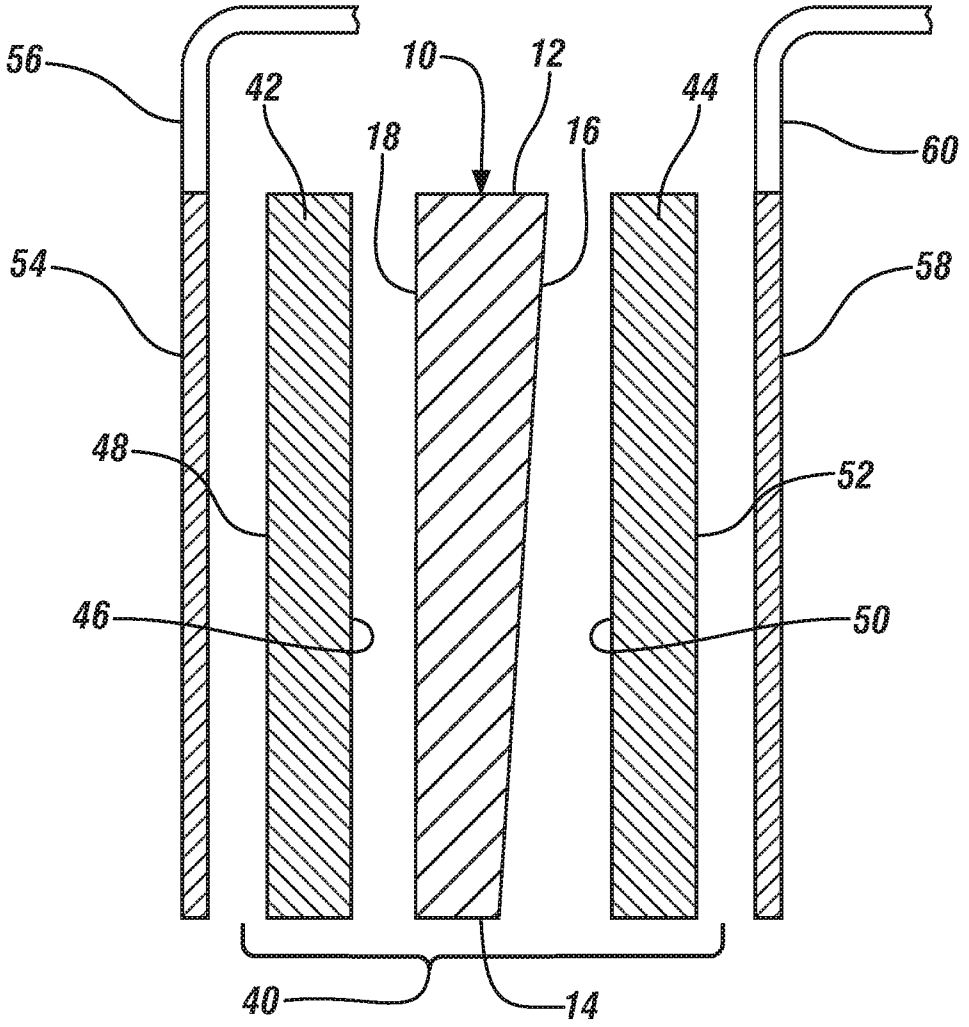


FIG. 5

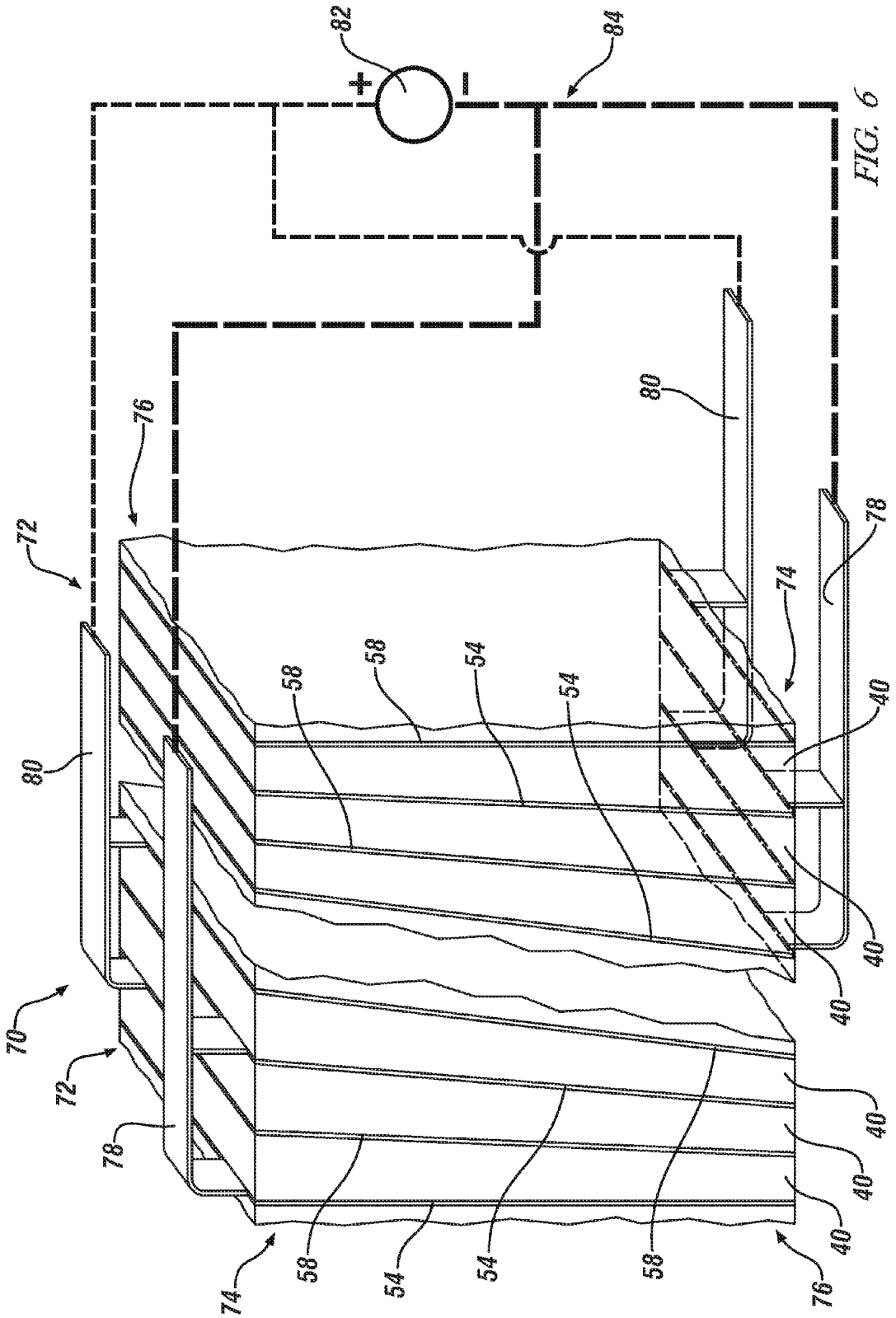


FIG. 6

POROUS POLYMER SEPARATOR LAYER HAVING A NON-UNIFORM CROSS-SECTIONAL THICKNESS FOR USE IN A SECONDARY LIQUID-ELECTROLYTE BATTERY

TECHNICAL FIELD

The technical field of this disclosure relates generally to separators for use in secondary liquid-electrolyte batteries and, more specifically, to a porous polymer separator layer having a non-uniform cross-sectional thickness.

BACKGROUND

A secondary liquid-electrolyte battery generally contains at least one electrochemical battery cell that include a negative electrode, a positive electrode, and a separator situated between the electrodes. The negative and positive electrodes are constructed from materials that can participate in both oxidation and reduction reactions. Such materials allow an electric current to be reversibly passed between the electrodes, external to the separator, while an ionic species migrates through the separator within a liquid electrolyte to electrochemically balance the current flow. This type of electrical and ionic current flow occurs spontaneously (cell discharge phase in which oxidation occurs at the negative electrode and reduction occurs at the positive electrode) or is compelled (cell charge phase in which oxidation occurs at the positive electrode and reduction occurs at the negative electrode). The electric current generated during cell discharge may be used to power, at least in part, an electrical load while an applied voltage from an external power source may be used to charge, or re-power, the cell once its current capacity has fallen to an undesirable level.

The separator facilitates operation of the electrochemical battery cell by providing a porous and electrically-insulative mechanical support barrier between the two electrodes. The separator, in general, has a porosity sufficient to contain the liquid electrolyte—which can communicate the ionic species—yet is thermally, chemically, and mechanically stable enough to separate the negative and positive electrodes over the course of many discharge/charge cell cycles so that a short-circuit is prevented. A wide variety of materials, either alone or in combination with one another, have been either utilized or investigated for construction of the separator with the goal of imparting long term operational reliability to the separator within different working environments. The most commonly used separators today are made from a single flat polyolefin sheet membrane or a laminate of several flat polyolefin sheet membranes. The particular polyolefins usually employed are those derived from simple low-carbon number olefins, such as polypropylene and polyethylene.

The electrochemical battery cell, in order to interact with the electrical load and the external power source, is configured for connection to an external circuit that provides an electric current path between the negative and positive electrodes around the separator. Each of the negative and positive electrodes, for instance, is typically associated with a metallic current collector that helps distribute the electric current passing through the external circuit to and from all electrochemically active regions of the electrodes. A connection feature such as connector tab may be included on each of the metallic current collectors. The connection feature may protrude away from the electrochemical battery cell to operatively establish an electrical connection with the external circuit. This is usually accomplished by connecting the protruding connection features associated with the negative and positive electrodes to negative and positive terminals, respectively, in either a serial or parallel relationship with the connection features associated with other electrochemical battery cells. Negative and positive terminals may not be needed, however, if the secondary liquid-electrolyte battery includes only one electrochemical battery cell.

The connection feature included on a metallic current collector is commonly located near a peripheral edge of the current collector for various practical reasons including, among others, accessibility. But locating the connection feature in this way can cause an uneven current density distribution to develop within the electrochemical battery cell. For instance, during oxidation of the associated electrode, the connection feature may have a tendency to draw electric current from a nearby portion of the electrode at a greater rate than more distant portions. Likewise, during reduction of the associated electrode, the connection feature may have a tendency to make electric current more readily available for ionic species reduction at a nearby portion of the electrode as opposed to other portions further removed. Such differences in electrochemical activity can become even more pronounced if the connection features of the metallic current collectors are all located on the same side of the cell. Nonetheless, however developed, a region of the electrochemical battery cell that experiences disparately greater current density can result in some potentially undesirable effects over the life of the battery.

SUMMARY OF THE DISCLOSURE

A porous polymer separator layer that may function as all or part of a separator for an electrochemical battery cell (sometimes referred to as "a cell" or "the cell" for brevity) of a secondary liquid-electrolyte battery, such as a lithium ion battery, and a method of making the same are disclosed. The porous polymer separator layer includes a first edge and a second edge that define a width of the separator layer, and a first major face and a second major face extending between the first and second edges that define a thickness of the separator layer. The first edge has a first height and the second edge has a second height. Each of the first and second heights establish the thickness of the porous polymer separator layer at the first and second edges, respectively, and together provide the separator layer with a non-uniform cross-sectional thickness spanning the width of the separator layer. To establish this non-uniform cross-sectional thickness, for example, the first height of the first edge is rendered greater than the second height of the second edge so that the thickness of the porous polymer separator layer decreases from the first edge to the second edge as the major faces converge towards one another.

The structure of the porous polymer separator layer is provided by a polymer matrix comprised of a polyolefin. The polymer matrix includes a porosity that traverses the thickness of the porous polymer separator layer and connects the opposed first and second major faces. Such a porosity is derived from pores that are distributed throughout the polymer matrix. These pores are generally small enough and sufficiently distributed that contact between the electrode surfaces situated on opposite sides the porous polymer separator layer can be prevented yet still large enough and adequately interconnected to contain a liquid electrolyte and facilitate ion transport. The porosity of the porous polymer separator layer as a whole is usually anywhere between about 30% and about 90%. The porosity may be distributed to allow communication of an ionic species through the thickness of

the porous polymer separator layer coextensively between the first and second opposed major faces when an infiltrated liquid electrolyte is present. In a preferred embodiment, moreover, the porosity is evenly distributed throughout the porous polymer separator layer.

The porous polymer separator layer, when incorporated into an electrochemical battery cell of a secondary liquid-electrolyte battery, may be oriented in a way that helps maintain a more evenly distributed current density within the cell. To accomplish this objective, for example, the porous polymer separator layer may be oriented so that the first edge is located between confronting surfaces of a negative electrode and a positive electrode in a region of the cell where the electrodes are prone to experience increased relative electrochemical activity. Orienting the first edge of the porous polymer separator layer in this way compensates for the tendency to experience increased localized electrochemical activity in that particular region of the cell by making the mobile ionic species travel a greater distance through the thicker edge of the separator layer. The electrochemical activity experienced throughout the electrodes, and thus the current density of the electrochemical battery cell, is therefore more equalized since the porous polymer separator layer renders the ionic species more evenly available to the electrodes despite differing electrochemical driving forces.

The porous polymer separator layer may be made by a process in which the separator layer is derived from a film having a non-uniform cross-sectional thickness similar to that sought to be imparted to the resultant separator layer. This process may involve (1) forming a film that has a cross-section with a non-uniform thickness from a polymer composition and (2) deriving the porous polymer separator layer from the film. Deriving the porous polymer separator layer may be accomplished according to a "dry" working process or a "wet" working process. In the dry working process, the polymer composition includes a polyolefin and, optionally, particles having a melting temperature equal to or greater than the polyolefin. The polymer composition is heated to melt the polyolefin, then formed into the film, and followed by annealing of the film and then uniaxial stretching of the film to form the porous polymer separator layer. In the wet working process, the polymer composition includes a polyolefin, a plasticizer absorbed into the polyolefin, and, optionally, particles having a melting temperature equal to or greater than the polyolefin. The polymer composition is heated to melt the polyolefin, then and formed into the film, and followed by biaxial stretching of the film and extraction of the plasticizer from the film to form the porous polymer separator layer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a generalized perspective view of a porous polymer separator layer for use in an electrochemical battery cell of a secondary liquid-electrolyte battery.

FIG. **2** is a generalized cross-sectional view of the porous polymer separator layer shown in FIG. **1** taken along the line **2-2**.

FIG. **3** is a generalized and partially magnified cross-sectional view of the porous polymer separator layer taken from the encircled portion of FIG. **2** and identified by the numeral **3**.

FIG. **4** is a general and diagrammatic depiction of a dry working process (left side) and a wet working process (right side) that may be used to make the porous polymer separator layer shown in FIGS. **1-3**.

FIG. **5** is a generalized cross-sectional view of an electrochemical battery cell for a lithium ion battery that includes a negative electrode, a positive electrode, and the porous polymer separator layer depicted in FIGS. **1-3** disposed between the negative and positive electrodes.

FIG. **6** is a generalized and schematic illustration of a lithium ion battery that includes several of the electrochemical battery cells depicted in FIG. **5**.

DETAILED DESCRIPTION

A preferred exemplary embodiment of a porous polymer separator layer **10** having a non-uniform cross-sectional thickness, and a method of making the separator layer **10** by a working process, are disclosed. The porous polymer separator layer **10** may be used in an electrochemical battery cell of a secondary liquid-electrolyte battery. An electrochemical battery cell of this type generally includes a negative electrode, a positive electrode, and a separator disposed between, and in interfacial contact with, the two electrodes. It further satisfies two general operational requirements: first, the electrochemical half-reactions that take place at the negative and positive electrodes are reversible so that the cell's capacity can be discharged multiple times and, second, the separator situated between the negative and positive electrodes is infiltrated with a liquid electrolyte that can communicate an electrochemically appropriate ionic species between the electrodes. Some notable examples of a secondary liquid-electrolyte battery that contains at least one such electrochemical battery cell—and usually a plurality of such cells electrically coupled together depending on the expected use of the battery—are a lithium ion battery, a nickel-cadmium battery, a nickel-metal hydride battery, and a lead-acid battery.

The separator is included in the electrochemical battery cell to physically separate and electrically insulate the electrodes while permitting the internal migration of a mobile ionic species through its infiltrated liquid electrolyte. The porous polymer separator layer **10** disclosed here is mechanically and thermally stable enough, and is sufficiently porous, that it may function entirely as a single-layer separator or it may be combined with other materials to fabricate a multi-component composite separator. The porous polymer separator layer **10**, on account of its non-uniform cross-sectional thickness, can also be oriented relative to the negative and positive electrodes to achieve a more evenly distributed current density within the cell. This may be accomplished by positioning the thicker portion of the porous polymer separator layer **10** between the surfaces of the negative and positive electrodes at a region of the cell where the electrodes have a tendency to experience more electrochemical activity because of some driving force such as, for example, a close proximity to a connection feature of an associated metallic current collector.

The porous polymer separator layer **10** will now be described in more detail within the context of an electrochemical battery cell for a lithium ion battery. This type of battery technology is considered by the automobile industry to be one of the primary candidates for integration into the powertrains of a hybrid electric vehicle (HEV) and an extended range electric vehicle (EREV), as well as other possible applications. Skilled artisans will, however, be more than capable of adapting this disclosure to the other types of secondary liquid-electrolyte batteries mentioned before. This is because the use of the porous polymer separator layer **10** in an electrochemical battery cell of a lithium ion battery is generally representative of how the separator layer **10** would be used in those other types of liquid-electrolyte batteries. The fact that each of the previously-mentioned liquid-elec-

trolyte batteries contains one or more electrochemical battery cells having different electrode constructions, different liquid electrolytes, and different electrochemical mechanisms is not considered to be a significant impediment to the robust use of the porous polymer separator layer **10**.

FIGS. **1-3** illustratively show an idealized depiction of the porous polymer separator layer **10** that may function as all or part of a separator for use in an electrochemical battery cell of a lithium ion battery. The porous polymer separator layer **10** includes a first edge **12** and a second edge **14** opposite the first edge **12**. These edges **12**, **14** define a width **200** of the porous polymer separator layer **10**. Extending between the first and second edges **12**, **14** opposite one another is a first major face **16** and a second major face **18**. These faces **16**, **18** define a thickness **300** of the porous polymer separator layer **10** in a direction generally perpendicular to the width **200** direction. The first and second major faces **16**, **18** are significantly larger in surface area than the first and second edges **12**, **14** because they are intended to emulate the size of the spaced-apart confronting surfaces of a negative electrode and a positive electrode which, in the construct of an electrochemical battery cell of a lithium ion battery, are situated on opposite sides of the separator layer **10**.

The porous polymer separator layer **10**, as shown best in FIG. **2**, has a cross-section spanning its width **200** that exhibits a non-uniform thickness. This cross-section includes the first edge **12**, which has a first height **400**, and the second edge **14**, which has a second height **500**. These first and second edge heights **400**, **500** establish the spacing between the first and second major faces **16**, **18** and, consequently, the thickness **300** of the porous polymer separator layer **10** at the first and second edges **12**, **14**, respectively. Moreover, in terms of their relationship, the height **400** of the first edge **12** is greater than the height **500** of the second edge **14** such that the thickness **300** of the porous polymer separator **10** decreases from the first edge **12** to the second edge **14** as the major faces **16**, **18** converge towards one another. The difference between the heights **400**, **500** of the first and second edges **12**, **14** is not a random variation that falls within an acceptable tolerance range usually associated with a uniform cross-sectional thickness; rather, it is meant and controlled to be more pronounced. The height **400** of the first edge **12**, more specifically, in a preferred embodiment, is about 2% to about 50% greater than the height **500** of the second edge **14**, more preferably about 5% to about 40% greater, and most preferably about 5% to about 30% greater.

The heights **400**, **500** of both the first and second edges **12**, **14** are preferably between about 5 μm and about 50 μm, more preferably between about 10 μm and about 45 μm, and most preferably between about 20 μm and about 35 μm while maintaining their differential height relationship. For example, if the height **500** of the second edge **14** is 20 μm, then the height **400** of the first edge **14** is preferably between about 20.4 μm (2% greater) to about 30 μm (50% greater), more preferably between about 21 μm (5% greater) to about 28 μm (40% greater), and most preferably between about 21 μm (5% greater) to about 26 μm (30% greater). The exact height difference sought to be achieved between the first and second heights **400**, **500** for a given lithium ion battery application, however, depends on several factors. Considerations like the composition of the porous polymer separator layer **10**, the composition and construction of the negative and positive electrodes, and the surface areas of the first and second major faces **16**, **18**, to name but a few factors, would all be considered by a skilled artisan when selecting a target height **400**, **500** for the first and second edges **12**, **14** and the corresponding non-uniform thickness at the cross-section of the porous polymer separator layer **10**.

The bulk structure of the porous polymer separator layer **10** is provided by a polymer matrix **20**, as depicted generally in FIG. **3**. The polymer matrix **20** is comprised of a polyolefin. This type of polymer is derived at least in part from olefinic monomers that encompass a variety of acyclic hydrocarbon compounds having a reactive carbon-carbon double bond (i.e., —C═C). The polyolefin designation as used here broadly includes a polyolefin homopolymer, a polyolefin-based heteropolymer (i.e., a copolymer), or a blend of more than one type of polyolefin constituent. Some specific examples of the polyolefin include, for example, polyethylene (PE) including any of its variations (such as HDPE, LDPE, LLDPE, and UHMWPE), polypropylene (PP), or a blend of PE and PP. But the most preferred composition of the polymer matrix **20** is PE or PP in part because of the mechanical and chemical stability, acceptable cost, and ease of manufacture of those two polyolefinic materials.

In order to accommodate the internal communication of lithium ions, a porosity that traverses the thickness **300** of the porous polymer separator layer **10** and connects the opposed first and second major faces **16**, **18** is provided within polymer matrix **20**. The term "porosity" as used here refers to the volume percentage of the porous polymer separator **10** within the polymer matrix **20** that is occupied by pores **22**, which are depicted generally in FIG. **3** although not necessarily to scale. These pores **22** may vary in size, shape, and spacing. But collectively they are generally small enough and sufficiently distributed to prevent contact between the confronting surfaces of the negative and positive electrodes the separator layer **10** is meant to separate yet still large enough and adequately interconnected to contain a liquid electrolyte and facilitate lithium ion mobility. A notable proportion of the pores, for instance, preferably has at least one cross-sectional dimension sized between 0.01 to 2.0 μm, although such a size range is not exclusive or even necessary to ensure proper operability of the porous polymer separator layer **10**. Larger and smaller cross-sectional pore sizes may be present depending on the thickness **300** of the porous polymer separator layer **10**, the particular lithium ion battery application in which the porous polymer separator layer **10** is employed, and various other considerations known to skilled artisans.

The porosity of the porous polymer separator layer **10** as a whole is preferably anywhere between about 30% and about 90% and most preferably between about 40% and 70%. The porosity may be distributed such that lithium ions can be communicated through the thickness **300** of the porous polymer separator layer **10** coextensively between the first and second opposed major faces **16**, **18** when the separator layer **10** is infiltrated with a liquid electrolyte. The porosity may further be evenly distributed throughout the porous polymer separator layer **10**. An evenly distributed porosity would be present if, referring for the moment back to FIG. **2**, the cross-section taken across the width **200** of the porous polymer separator layer **10** is divided vertically along the thickness **300** direction into two segments of equal width, one that includes the first edge **12** (segment **1**) and one that includes the second edge **14** (segment **2**), and the porosity of those segments as measured by any suitable technique, such as mercury porosimetry, deviates from the porosity of the porous polymer separator layer **10** as a whole by no more than 3%. For example, if the entire porous polymer separator layer **10** has a porosity of 60%, and a segment **1** and a segment **2** of the porous polymer separator layer **10** are obtained as just described, and each of those segments has a porosity within

the range of 57% to 63%, then the porous polymer separator layer **10** can be said to have an evenly distributed porosity.

One or more types of particles **24** having a melting temperature equal to or higher than the melting temperature of the polyolefin that constitutes the polymer matrix **20** may be dispersed, if desired, within the polymer matrix **20**, as depicted generally in FIG. **3**. The presence of such particles **24** may enhance the mechanical properties of the porous polymer separator layer **10** and help mitigate or altogether prevent thermal shrinkage. Some particular types of particles **24** that may be employed include ceramic particles, cross-linked thermoset polymer particles, and engineering thermoplastic polymer particles, to name but a few examples. Exactly how much of the particles **24** are dispersed within the polymer matrix **20** is subject to a wide variance. In many instances, however, the particles **24** are preferably present in an amount that ranges from about 0 wt. % to about 400 wt. % of the polymer matrix **20**; that is, the porous polymer separator layer **10** may comprise, by weight, up to about 80% of the dispersed particles **24** based on the total weight of the polymer matrix **20** and the particles **24**. A preferred particle diameter for the particles **24** is about 0.005 μm to about 15 μm and, most preferably, from about 0.05 μm to about 3 μm.

Many specific materials may be employed as the optionally-dispersed particles **24**. Some examples of suitable ceramic particles include those of alumina, titania, silica, ceria, zirconia, silicon carbide, boron carbide, titanium nitride, silicon nitride, titanium silicide, tungsten silicide, aluminum boride, titanium boride, mullite, spodumene, zirconium silicate, sillimanite, petalite, and mixtures thereof. Some examples of suitable cross-linked thermoset polymer particles include the cured forms of polyimides, polyurethanes, phenol-formaldehyde resins, melamine-formaldehyde resins, epoxy resins, and mixtures thereof. Some examples of suitable engineering thermoplastic polymer particles include those of a polyimide such as polyetherimide, a polyketone such polyether ketone (PEK), polyether ether ketone (PEEK), and polyarylether ketone (PAEK), a polysulfone such as standard polysulfone, polyarylsulfone, polyethersulfone, and polyphenylsulfone, polyethylene terephthalate, polyphenylene sulfide, ethylene-vinyl acetate, and mixtures thereof. Of course other types of materials not specifically listed above may also be used to fabricate the particles **24** as is generally understood by skilled artisans.

The porous polymer separator layer **10** may be made by a working process shown generally and diagrammatically in FIG. **4**. This process includes (1) forming a film **110**, which has a cross-section with a non-uniform thickness, from a polymer composition and (2) deriving the porous polymer separator layer **10** with its ascribed porosity from the film **110**. Deriving the porous polymer separator layer **10** may be accomplished according to a "dry" working process or a "wet" working process depending on the preference of a skilled artisan. Each of these processes can achieve the desired porosity of the porous polymer separator layer **10** as described above albeit with slightly different structural morphologies. Making the porous polymer separator layer **10** by way of the dry working process, for instance, produces relatively small slit-like pores within the polymer matrix **20** and provides the separator layer **10** with anisotropic mechanical properties. Employing the wet working process, on the other hand, produces more proportionately-sized pores and provides the separator layer **10** with more isotropic mechanical properties. The "dry" and "wet" working processes will be described below in further detail. Additional processing may be performed after the porous polymer separator layer **10** is made such as, for example, cutting or trimming or otherwise shaping the separator layer **10** in light of its expected use, as well as combining the separator layer **10** with another material or layer to form a multi-component composite separator if desired.

The polymer composition used to make the film **110** includes the polyolefin selected to fabricate the polymer matrix **20**. Selection of the particular polymer material may depend on several factors including its mechanical strength, temperature resistance, chemical resistance, cost, availability, and ease of manufacturing. The polymer composition may further include a plasticizer if the wet working process is used to make the porous polymer separator layer **10**. Other materials may also be included in the polymer composition including the particles **24** having a melting temperature equal to or higher than the melting temperature of the polyolefin that comprises the polymer matrix **20**. Additives such as viscosity modifiers, antioxidants, antistatic agents, and other processing aids may also be included in the polymer composition especially if the wet working process is utilized.

The amounts of the several materials that may be included in the polymer composition are determined based on several factors such as the intended thickness, porosity, and mechanical properties of the porous polymer separator layer **10**. But in general they typically fall within the ranges of about 20 wt. % to about 100 wt. % of the polyolefin based on the weight of the polymer composition if no plasticizer is present, about 20 wt. % to about 95 wt. % of the polyolefin based on the weight of the polymer composition if a plasticizer is present, about 5 wt. % to about 70 wt. % of the plasticizer, if included, based on the total weight of the polyolefin and the plasticizer, and about 10 wt. % to about 400 wt. % of the particles **24**, if included, based on the weight of the polyolefin included in the polymer composition. Of course the exact amounts of each material used can be determined by as skilled artisan and adjusted accordingly, as needed, to achieve the desired structure and properties of the porous polymer separator layer **10**.

The polymer composition may be heated and mechanically worked to melt the polyolefin and disperse the optionally-included particles **24**. Melting the polyolefin allows the polymer composition to be formed into the film **110** with relative ease. The polymer composition, when heated, may include a melt of the polyolefin (hereafter referred to as a "polyolefin melt") or a melt mixture of the polyolefin (hereafter referred to as a "polyolefin melt mixture") during formation into the film **110**. A polyolefin melt refers to embodiments in which the melted polyolefin is not mixed with a plasticizer. This type of polymer composition is generally used during practice of the dry working process. Conversely, a polyolefin melt mixture refers to embodiments in which the melted polyolefin is mixed with a plasticizer. This type of polymer composition is generally used during practice of the wet working process. The temperature of the polymer composition that needs to be achieved to maintain the polyolefin melt or melt-mixture during formation of the film **110** may vary. But in most instances a temperature between about 90° C. to about 180° C. is sufficient during formation of the film **110** to ensure proper flow and formability.

The film **110** of the polymer composition may be formed by any suitable procedure notwithstanding the contents of the polymer composition or the stretching technique employed. The film **110** is formed to have the same basic shape as the resultant porous polymer separator layer **10** albeit with a greater thickness **300'** dimension and possibly a lesser width **200'** dimension to accommodate for the effects of subsequent stretching. In other words, the film **110** has a first edge **112**, a second edge **114** opposite the first edge **112**, and opposed first and second major faces **116**, **118** extending between the first

and second edges **112**, **114**. The first and second edges **112**, **114** define a width **200'** of the film **110** and the first and second major faces **116**, **118** define a thickness **300'** of the film **110** generally perpendicular to the width **200'** direction. Still further, the first edge **112** has a height **400'** that is greater than a height **500'** of the second edge **114**, so that, like the porous polymer separator layer **10**, a cross-section of the film **110** spanning its width **200'** exhibits a non-uniform thickness. Each height **400'**, **500'** dimension is typically about 20% to about 200% greater than the desired corresponding height **400**, **500** dimension of the porous polymer separator layer **10**. The width **200'** dimension of the film **110** may be equal to the width **200** dimension of the porous polymer separator layer **10** or, in some embodiments, less than the width **200** dimension of the porous polymer separator layer **10** by about 5% to about 50%.

The film **110** is preferably formed by a film extrusion procedure. Such a procedure generally involves pushing the polymer composition through a non-uniform opening of an extruder die at a temperature sufficient to at least temporarily maintain the polyolefin in melt or melt-mixed form (i.e., between about 90° C. to about 180° C.). The film **110** is urged through the opening of the extrusion die along an extrusion direction (sometimes referred to as the machine direction) with the width **200'** of the film **110** being oriented transverse to the extrusion direction. A roller or a succession of rollers (not shown) may receive the film **110** as it emerges from the extruder die. The roller(s) may be configured to quench the film **110** as is generally well understood in the art. Other procedures may of course be used to produce the film **110** besides film extrusion. Procedures such as blown extrusion, slot-die coating, comma-bar coating, and other procedures not specifically mentioned here but known to skilled artisans, may also be employed to form the film **110**.

Deriving the porous polymer separator layer **10** from the film **110** usually involves stretching the film **110** according to known techniques. The film **110** may be stretched either uniaxially or biaxially. Such stretching causes the thickness **300'** of the film **110** to decrease—and the width **200'** of the film **110** to increase if stretching is practiced in the width **200'** direction—and also helps cultivate the porosity ultimately provided in the separator layer **10**. In other words, the first edge **12**, the second edge **14**, the first major face **16**, and the second major face **18** of the porous polymer separator layer **10** are derived, respectively, from the first edge **112**, the second edge **114**, the first major face **116**, and the second major face **118** of the film **110**. The height relationship originally devised between the first and second edges **112**, **114** of the film **110** is also maintained during stretching of the film **110** such that the height **400** of the first edge **12** of the porous polymer separator layer **10** is greater than the height **500** of the second edge **14** as previously described. The intricacies of both the "dry" and "wet" working processes will now be described in more detail.

Dry Process

In the dry process, which is depicted generally and diagrammatically on the left side of FIG. **4**, the polymer composition comprises a polyolefin selected to constitute the polymer matrix **20** of the porous polymer separator layer **10** but does not include a plasticizer. Heating the polymer composition and melting the polyolefin thus produces a polyolefin melt. The particles **24** may be included in the polymer composition, if desired, for suspension in the polyolefin melt during formation of the film **110** and eventual dispersal within the polymer matrix **20**. The heated polymer composition is preferably formed into the film **110** with a non-uniform cross-sectional thickness and a width **200'** approximately equal to the intended width **200** of the porous polymer separator layer **10** being made as has been previously described. The film **110** is then cooled in the ambient environment to allow the polyolefin melt to solidify and crystallize. Such crystallization of the polyolefin melt forms ordered lamellae crystallite regions bordered by amorphous phase regions throughout the film **110** within the solidified polyolefin. The film **110** is considered "dry" at this point since the polyolefin is solidified and no plasticizer is present.

The film **110** is then annealed to further crystallize the polyolefin. Annealing is preferably performed by heating the film **110** to a temperature within 50° C. of the melting point of the polyolefin for a suitable length of time (represented by the arrows identified as **600**). This typically entails heating the film **110** to somewhere in the range of 70° C. to about 140° C. in an oven or other suitable heating contraption for a few seconds up to several tens of minutes depending on the particular polyolefin present and the thickness of the film **110**. The additional heat treatment associated with annealing produces more lamellae crystallite regions within the polyolefin and reinforces the ones formed during cooling of the polyolefin melt. The many lamellae crystallite regions situated throughout the polyolefin establish boundaries with the amorphous phase regions. These boundaries eventually become the pores **22** in the polymer matrix **20** following stretching of the film **110** into the porous polymer separator layer **10**.

The film **110** is then uniaxially stretched in a direction perpendicular to its width **200'** (represented by the arrows identified as **700**) to form the porous polymer separator layer **10**. If the film **110** has been formed by extrusion, this stretching would be performed in the extrusion direction. Stretching the film **110** in this way pulls apart the lamellae crystallite regions and the surrounding amorphous phase regions present throughout the polyolefin and forms the polymer matrix **20** with an internal distribution of the pores **22** that provide the porous polymer separator layer **10** with its porosity. A similar mechanism occurs around the particles **24** contained in the film **110** during stretching to further enhance the porosity of the separator layer **10** if those particles **24** are indeed present. The film **110** may also be stretched in a direction parallel to the width W' of the film **110**, if desired, to try and further affect the size and shapes of the pores **22**. But such stretching is typically omitted since it tends to have little impact the porosity of the porous polymer separator layer **10** because of the way the lamellae crystallite regions are oriented.

The uniaxial stretching of the film **110** maintains the height relationship between the first and second edges **112**, **114** since the film **110** is stretched perpendicular to its width **200'** in a fairly uniform fashion. Moreover, to help stretch the film **110** uniformly, heat may be applied to the first edge **112** of the film **110** in a greater amount than the second edge **114** to establish a temperature gradient across the width **200'** of the film **110**. A temperature gradient of this type can equalize the modulus of elasticity—which is inversely proportional to temperature—of the first and second edges **112**, **114** so that the heights **400'**, **500'** of these edges **112**, **114** are reduced by substantially the same amount during stretching of the film **110**. The heat applied to the first edge **112** of the film **110** is preferably modest enough that the temperature in that area of the film **110** remains lower than the annealing temperature so that localized annealing of only part of the film **110** is avoided.

Wet Process

In the wet process, which is depicted generally and diagrammatically on the right side of FIG. **4**, the polymer com-

position comprises a polyolefin selected to constitute the polymer matrix **20** of the porous polymer separator layer **10** and a plasticizer. The plasticizer included in the polymer composition is a low volatility organic liquid that can be absorbed into the polyolefin and, once absorbed, lowers the glass transition temperature of the polyolefin. The addition of the plasticizer therefore generally provides the polyolefin with enhanced flexibility, resiliency, and melt flow capabilities. Any plasticizer that is compatible with the polyolefin may be used. One such plasticizer is mineral oil. Mineral oil is comprised of a mixture of heavy alkanes derived from petroleum or some other non-vegetable source. At least 80% of the alkanes are preferably in the C15-C40 range and the boiling point of the mineral oil is greater than about 320° C. and typically greater than 350° C. A specific type of mineral oil that is preferably employed is paraffin oil. Paraffin oil is contains at least 80% of n-alkanes in the C15-C40 range and is generally commercially available from numerous suppliers including, for example, Chevron and Exxon Mobile. Various additives may also be included into the polymer composition.

Heating the polymer composition and melting the polyolefin thus produces a polyolefin melt mixture. The particles **24** may be included in the polymer composition, if desired, for suspension in the polyolefin melt mixture during formation of the film **110** and eventual dispersal within the polymer matrix **20**. The heated polymer composition is preferably formed into the film **110** with a non-uniform cross-sectional thickness and a width **200'** less than the intended width **200** of the porous polymer separator layer **10** being made as has been previously described. The film **110** is then cooled in the ambient environment to allow the polyolefin melt mixture to gelatinize. Such gelatinization results in the absorbed plasticizer being trapped throughout the film **110** within the gelled polyolefin as small liquid confinements that occupy future pore sites. The film **110** is considered "wet" at this point since the polyolefin is gel-like in appearance due to the presence of the plasticizer.

After being formed, the film **110** may be stretched biaxially in a direction parallel to the width **200'** of the film **110** (represented by the arrows identified as **720**) and a direction perpendicular to the width **200'** (represented by the arrows identified as **730**) of the film **110**. If the film **110** has been formed by extrusion, this stretching would be performed in both the extrusion and transverse directions. Stretching the film **110** in this way enlarges the future pore sites occupied by the plasticizer within the gelled polyolefin. A similar mechanism occurs around the particles **24** contained in the film **110** during stretching to further enhance the porosity of the separator layer **10** if those particles **24** are indeed present. Biaxial stretching is more constructive here than in the dry working process because ordered lamellae crystallite regions that can restrict stretching in a direction parallel to the width **200'** of the film **110** are generally not produced in the gelled polyolefin when the film **110** is formed.

The biaxial stretching of the film **110** maintains the height relationship between the first and second edges **112**, **114** since the film **110** is stretched both parallel and perpendicular to its width **200'** in a fairly uniform fashion. Moreover, to help stretch the film **110** uniformly, heat may be applied to the first edge **112** of the film **110** in a greater amount than the second edge **114** to establish a temperature gradient across the width **200'** of the film **110**. A temperature gradient of this type can equalize the modulus of elasticity—which is inversely proportional to temperature—of the first and second edges **112**, **114** so that the heights **400'**, **500'** of these edges **112**, **114** are reduced by substantially the same amount during stretching of the film **110**. The heat applied to the first edge **112** of the film **110**, like in the dry working process, is preferably modest enough that the temperature in that area of the film **110** remains lower than the annealing temperature so that localized annealing of only part of the film **110** is avoided.

The plasticizer contained in the film **110** is then extracted to form the porous polymer separator layer **110**. The plasticizer may be extracted by washing the film **110** with solvent that is miscible with, and more volatile than, the plasticizer but does not dissolve or absorb into the polyolefin. Such washing may involve immersing the film **110** into a bath that includes the volatile solvent or some other suitable procedure. The presence of the volatile solvent (represented by the arrows identified as **800**) in an around the film **110** during washing draws the plasticizer (represented by the arrows identified as **900**) out of the polyolefin and leaves behind the polymer matrix **20** with an internal distribution of the pores **22** that provide the porous polymer separator layer **10** with its porosity. Very little if any of the volatile solvent usually remains within the porous polymer separator layer **10** after the separator layer **10** is removed from the bath or at the conclusion of whatever washing procedure is performed. But to be sure the porous polymer separator layer **10** may be optionally heated to remove any residual volatile solvent that may still be present. A preferred volatile solvent that may be used to extract the plasticizer from the film **110** is methylene chloride (CH_2Cl_2).

It should be noted that the order in which stretching of the film **110** and extraction of the plasticizer from the film **110** are performed in the exemplary wet working embodiment just described may be switched. That is, after the film **110** is formed, the plasticizer may be extracted from the film **110** followed by biaxial stretching of the film **110** to form the porous polymer separator layer **10**. The order in which these two acts are performed may have some impact the structure or function of the porous polymer separator layer **10**. Stretching the film **110** before extracting the plasticizer may result in more proportionately-sized pores than when those two acts are reversed. Both ways of practicing the stretching and extracting acts are nonetheless fully acceptable in the fabrication of the porous polymer separator layer **10** and, in some instances, are implemented in one way or the other based on manufacturing practicalities as opposed to some performance-related reason.

Referring now back to FIGS. **1-3**, the porous polymer separator layer **10** may be used in a way that helps maintain a more evenly distributed current density in an electrochemical battery cell of a lithium ion battery. To be used for such a purpose, for example, the porous polymer separator layer **10** may be oriented within the cell such that the first edge **12** is located between confronting surfaces of the negative and positive electrodes in a region of the cell where the electrodes are prone to experience increased relative electrochemical activity. Oftentimes this region coincides with a side of the electrochemical battery cell proximate a connection feature of a metallic current collector associated with the negative electrode and/or a connection feature of a metallic current collector associated with the positive electrode. Orienting the first edge **12** of the porous polymer separator layer **10** in this way compensates for the tendency to experience increased localized electrochemical activity in that particular region of the cell by making the lithium ions travel a greater distance through the first edge **12** of the separator layer **10** relative to the second edge **14**. In other words, because of their height differences, the first edge **12** of the porous polymer separator layer **10** provides a greater overall resistance to lithium ion transport between the electrodes than the second edge **14** and, as such, orienting those edges **12**, **14** at regions of the cell having expectations of greater and lesser relative electro-

chemical activities, respectively, helps equalize the electrochemical activity between the electrodes throughout the cell.

The ability of the porous polymer separator layer **10** to help maintain a more evenly distributed current density can enhance the longevity and reliability of the electrochemical battery cell in some instances. One potential benefit of a more evenly distributed current density is the better utilization of the electrodes; that is, the portions of the electrodes closer to the connection features of the metallic current collectors will be less likely to decay at an accelerated rate while the portions more removed from the connection features will be less likely to be operationally underused. Another potential benefit of a more evenly distributed current density is a more even temperature distribution within the cell. The avoidance of localized high temperature spots due to increased relative electrochemical activity can help maintain the integrity of the separator which, in addition to the porous polymer separator layer **10**, may include one or more polymer components that are susceptible to degradation at an elevated temperature. Yet another potential benefit of a more evenly distributed current density is better dimensional stability of the cell. For instance, the electrochemical battery cell may expand during the charging phase and contract during the discharge phase. A more evenly distributed current density can help achieve more uniform expansion and contraction of the electrodes and, as a result, reduce the occurrence of mechanical strains and corresponding stresses that can lead to loss of contact between the electrodes and the interadjacent separator.

The porous polymer separator layer **10** may be assembled into an electrochemical battery cell **40** of a lithium ion battery as depicted, for example, in the exploded cross-sectional view of FIG. **5**. The electrochemical battery cell **40** includes a negative electrode **42**, a positive electrode **44**, and the porous polymer separator layer **10** disposed between the two electrodes **42**, **44**. The negative electrode **42** includes an inner face surface **46** and an outer face surface **48** relative to the location of the porous polymer separator layer **10**. The positive electrode **44** similarly includes an inner face surface **50** and an outer face surface **52**. And when assembled into the electrochemical battery cell **40**, the inner face surfaces **46**, **50** of the negative and positive electrodes **42**, **44** confront one another and are pressed against the second major face **18** and the first major face **16** of the porous polymer separator layer **10**, respectively, as shown. Of course other arrangements of the porous polymer separator layer **10** are possible. For instance, the porous polymer separator layer **10** could be arranged so that the inner face surface **46** of the negative electrode **42** is pressed against the first major face **16** of the separator layer **10** and the inner face surface **50** of the positive electrode **44** is pressed against the second major face **18**. The porous polymer separator layer **10** may also be combined with other materials or layers to form a multi-component composite separator in which one or both of the first and second major faces **16**, **18** is no longer pressed against their respective electrode inner face surfaces **46**, **50**.

Situated on each side of the electrochemical battery cell **40** is a metallic current collector that includes a connector tab for accommodating an electrical connection. More specifically, a negative-side metallic current collector **54** that includes a projecting negative polarity tab **56** is associated with the negative electrode **42** and a positive-side metallic current collector **58** that includes a projecting positive polarity tab **60** is associated with the positive electrode **44**. The negative-side metallic current collector **54** supports the outer face surface **48** of the negative electrode **42** and the positive-side metallic current collector **50** supports the outer face surface **52** of the positive electrode **44**. The metallic current collectors **54**, **58** also preferably engage their respective electrode outer face surfaces **48**, **52** over an appreciable interfacial surface area, as shown, to facilitate the efficient collection and distribution of free electrons. Each of the metallic current collectors **54**, **58** may be constructed from any of a wide variety of electrically conductive metal. In a preferred embodiment, however, the negative-side metallic current collector **54** is a thin-film copper foil and the positive-side metallic current collector **58** is a thin-film aluminum foil.

A typical thickness of the electrochemical battery cell **40** extending between the metallic current collectors **54**, **58** is about 70 μm to about 350 μm. Each electrode **42**, **44** is preferably about 30 μm to 150 μm thick and, as already mentioned, the porous polymer separator layer **10** is about 10 μm to 50 μm thick at each of the first and second edges **12**, **14** with the first edge **12** being at least 2% thicker than the second edge **14**. The metallic current collectors **54**, **58** are usually about 5 μm to 20 μm thick. The relatively thin and flexible nature of the electrochemical battery cell **40** and its associated metallic current collectors **54**, **58** allows them to be rolled, folded, bent, or otherwise maneuvered into a variety of configurations depending on design specifications and spatial constraints. The electrochemical battery cell **40** can be configured, as shown in FIGS. **5** and **6**, for modular side-by-side stacking with other similarly constructed electrochemical battery cells or, in another embodiment, it may be rolled into a wound cylindrical configuration with other cells.

The negative electrode **42** is preferably constructed from a lithium host material such as, for example, graphite, silicon, or lithium titanate. The lithium host material may be intermingled with a polymeric binder material to provide the negative electrode with structural integrity and, optionally, a conductive fine particle diluent. The lithium host material is preferably graphite and the polymeric binder material is preferably one or more of polyvinyldiene fluoride (PVdF), an ethylene propylene diene monomer (EPDM) rubber, styrene butadiene rubber (SBR), a carboxymethoxy cellulose (CMC), polyacrylic acid, or mixtures thereof. Graphite is normally used to make the negative electrode because, on top of being relatively inert, its layered structure exhibits favorable lithium intercalation and deintercalation characteristics which help provide the electrochemical battery cell with a suitable energy density. Commercial forms of graphite that may be used to construct the negative electrode are available from Timcal Graphite and Carbon (headquartered in Bodio, Switzerland), Lonza Group (headquartered in Basel, Switzerland), and Superior Graphite (headquartered in Chicago, Ill.). The conductive particle diluent may be very fine particles of, for example, high-surface area carbon black.

The positive electrode **44** is preferably constructed from a lithium-based active material that stores intercalated lithium at a higher electrochemical potential than the lithium host material used to make the negative electrode. The same polymeric binder materials (PVdF, EPDM, SBR, CMC, polyacrylic acid) and conductive fine particle diluent (high-surface area carbon black) that may be used to construct the negative electrode may also be intermingled with the lithium-based active material for the same purposes. The lithium-based active material is preferably a layered lithium transition metal oxide, such as lithium cobalt oxide ($LiCoO_2$), a spinel lithium transition metal oxide, such as spinel lithium manganese oxide ($LiMn_2O_4$), a lithium polyanion, such as a nickel-manganese-cobalt oxide [$Li(Ni_XMn_YCO_Z)O_2$], lithium iron phosphate ($LiFePO_4$), or lithium fluorophosphate (Li_2FePO_4F). Some other suitable lithium-based active materials that may be employed include lithium nickel oxide ($LiNiO_2$), lithium aluminum manganese oxide ($Li_XAl_YMn_1$-

$_xO_2$), and lithium vanadium oxide (LiV_2O_5), to name but a few alternatives. Mixtures that include one or more of these recited lithium-based active materials may also be used to make the positive electrode.

The porous polymer separator layer **10** functions as an electrically insulative mechanical barrier that physically separates the confronting inner face surfaces **46**, **50** of the electrodes **42**, **44** to prevent a short-circuit in the electrochemical battery cell **40**. The porous polymer separator layer **10** is further infiltrated with a liquid electrolyte throughout its porosity that is able to communicate lithium ions. The liquid electrolyte, which also wets both electrodes **42**, **44**, is preferably a lithium salt dissolved in a non-aqueous solvent. Some suitable lithium salts that may be used to make the liquid electrolyte include $LiClO_4$, $LiAlCl_4$, LiI, LiBr, LiSCN, $LiBF_4$, $LiB(C_6H_5)_4$, $LiAsF_6$, $LiCF_3SO_3$, $LiN(CF_3SO_2)_2$, $LiPF_6$, and a mixture that includes one or more of these salts. The non-aqueous solvent in which the lithium salt is dissolved may be a cyclic carbonate (i.e., ethylene carbonate, propylene carbonate), an acyclic carbonate (i.e., dimethyl carbonate, diethyl carbonate, ethylmethylcarbonate), an aliphatic carboxylic ester (i.e., methyl formate, methyl acetate, methyl propionate), a γ-lactone (i.e., γ-butyrolactone, γ-valerolactone), an acyclic ether (i.e., 1,2-dimethoxyethane, 1,2-diethoxyethane, ethoxymethoxyethane), a cyclic ether (i.e., tetrahydrofuran, 2-methyltetrahydrofuran), or a mixture that includes one or more of these solvents.

The porous polymer separator layer **10** is also oriented between the electrodes **42**, **44** to help maintain a more uniform current density within the electrochemical battery cell **40**. The orientation of the porous polymer separator layer **10** shown here locates the first edge **12** between the confronting inner face surfaces **46**, **50** of the electrodes **42**, **44** in a region of the cell **40** adjacent the projecting polarity tabs **56**, **60** of the metallic current collectors **54**, **58** and, furthermore, locates the second edge **14** between the confronting inner face surfaces **46**, **50** of the electrodes **42**, **44** in a region of the cell **40** more removed from the projecting polarity tabs **56**, **60** than the first edge **12**. Such positioning of the first and second edges **12**, **14** of the porous polymer separator layer **10** compensates for the increased localized electrochemical activity that may occur in the cell **40** near the projecting polarity tabs **56**, **60**. This is because the lithium ions migrating between the electrodes **42**, **44** have to travel a greater distance through the porous polymer separator layer **10** at the first edge **12** relative to the second edge **14**. And making the lithium ions travel further through the porous polymer separator layer **10** in a region of the electrochemical battery cell **40** more prone to higher relative electrochemical activity helps make lithium ions more evenly available to the entire inner face surfaces **46**, **50** of the electrodes **42**, **44**.

The electrochemical battery cell **40** may be connected to a plurality of other similarly constructed electrochemical battery cells **40** to produce a lithium ion battery **70** as shown in FIG. **6**. Anywhere from five to fifty of the electrochemical battery cells **40** are typically connected in series (although a parallel connection is also permitted) to form a battery cell pack **72** that includes a thick end **74** and a thin end **76** in the stacking direction of the cells **10**. The thick end **74** of the cell pack **72** is the end that includes alignment of the first edges **12** of the porous polymer separators **10** and, conversely, the thin end **76** of the cell pack **72** is the end that includes alignment of the second edges **14**. Several of the battery cell packs **72** may be further connected in series or in parallel to provide the lithium ion battery **70** with the voltage and current capacity demanded for a particular vehicle application. A total of 20 to 150 electrochemical battery cells **40** are typically included in the lithium ion battery **70** if the battery **70** is intended to be used as an on-board a power source for a HEV or an EREV. To maintain an overall rectangular shape of the lithium ion battery **70**, moreover, a pair of adjacent battery cell packs **72** may be aligned with the thick end **74** of the cell pack **72** diametrically opposed to the thin end **76** of the other cell pack **72**.

A negative terminal **78** and the positive terminal **80** of the lithium ion battery **70** may be connected to an electrical device **82** as part of an external circuit **84** established between the negative electrodes **42** and the positive electrodes **44** of the several electrochemical battery cells **40**. The electrical device **82** may be a wide variety of electrical loads and power-generating devices. An electrical load is a power-consuming device that is powered fully or partially by the lithium ion battery **70**. Conversely, a power-generating device is one that charges or re-powers the lithium ion battery **70** through an applied external voltage. The electrical load and the power-generating device can be the same device in some instances. For example, the electrical device **82** may be an electric motor for a HEV or an EREV that is designed to draw an electric current from the lithium ion battery **70** during acceleration and provide a regenerative electric current to the lithium ion battery **70** during deceleration. The electrical load and the power-generating device can also be different devices. For example, the electrical load may be an electric motor for a HEV or an EREV and the power-generating device may be an AC wall outlet, an internal combustion engine, and/or a vehicle alternator.

The lithium ion battery **70** can provide a useful electrical current to the electrical device **82** by way of reversible electrochemical reactions that occur in the electrochemical battery cells **40** when the external circuit **84** is closed to connect the negative terminal **78** and the positive terminal **80** at a time when the negative electrodes **42** contain intercalated lithium (i.e., battery discharge). The electrochemical potential difference between the negative electrode **42** and the positive electrode **44** in each cell **40**—approximately 2.5 to 4.3V—drives the oxidation of intercalated lithium contained in the negative electrodes **42**. Free electrons produced by this oxidation reaction are collected by the negative-side current collectors **54** and are supplied to the negative terminal **78**. A flow of free electrons is harnessed and directed through the electrical device **82** from the negative terminal **78** to the positive terminal **80** and eventually to the positive electrodes **44** by way of the positive-side current collectors **58**. Lithium ions, which are also produced at the negative electrodes **42**, are concurrently carried through the porous polymer separator layers **10** by the liquid electrolyte in route to the positive electrodes **44** as well. The flow of free electrons through the electrical device **82** from the negative terminal **78** to the positive terminal **80** can be continuously or intermittently provided until the negative electrodes **42** are depleted of intercalated lithium or the capacity of the electrochemical battery cells **40** is otherwise spent.

The lithium ion battery **70** can be charged or re-powered at any time by applying an external voltage to the electrochemical battery cells **40** to reverse the electrochemical reactions that occur during discharge. The applied external voltage compels the otherwise non-spontaneous oxidation of intercalated lithium contained in the positive electrodes **44** to produce free electrons and lithium ions. The free electrons are collected by the positive-side current collectors **58** and are supplied to the positive terminal **80**. A flow of the free electrons is directed to the negative terminal **78** and eventually to the negative electrodes **42** by way of the negative-side current collectors **54**. The lithium ions are concurrently carried back through the porous polymer separator layers **10** by the liquid

electrolyte towards the negative electrodes **42** as well. The lithium ions and the free electrons eventually reunite and replenish the negative electrodes **42** with intercalated lithium to prepare the electrochemical battery cells **40** for another discharge phase. The external voltage may originate from the electrical device **82** as previously mentioned or by some other suitable mechanism.

While the lithium ion battery **70** shown in FIG. **6** includes several battery cell packs **72** arranged with diametrically opposed thick and thin ends **74**, **76**—with each of the cell packs **72** including several modularly-stacked electrochemical battery cells **40**—it should be understood the illustrated lithium ion battery **70** is a schematic representation of only one possible configuration. FIG. **6** is not intended to inform the relative sizes of the electrochemical battery cells' components or to limit the wide variety of structural configurations the lithium ion battery **70** may assume. Various structural modifications to the lithium ion battery **70** shown in FIG. **6** are possible, despite what is explicitly illustrated, as is well understood by skilled artisans. For example, the battery cell packs **72** could just as easily be arranged with their thick ends **74** adjacent to one another and their thin ends **76** adjacent to one another such that the lithium ion battery assumes a donut-shaped configuration instead of the rectangular configuration shown in FIG. **6**.

The above description of preferred exemplary embodiments is merely descriptive in nature and not intended to limit the scope of the claims that follow. Each of the terms used in the appended claims should be given its ordinary and customary meaning unless specifically and unambiguously stated otherwise in the specification.

The invention claimed is:

1. A method of making and using a porous polymer separator layer that may function as all or part of a separator for use in an electrochemical battery cell of a secondary liquid-electrolyte battery, the method comprising:

(a) providing a polymer composition that includes a polyolefin;

(b) forming a film having a cross-section with a non-uniform thickness from the polymer composition; and

(c) deriving a porous polymer separator layer from the film, the porous polymer separator layer having a cross-section that comprises a first edge and a second edge opposite the first edge, each of the first and second edges of the cross-section having a height that defines a thickness of the porous polymer separator layer at the first and second edges, respectively, and wherein the height of the first edge is greater than the height of the second edge such that the porous polymer separator layer has a thickness at the cross-section that decreases from the first edge to the second edge, the porous polymer separator layer further having a porosity in the range of about 30% to about 90% that is distributed evenly throughout the separator layer.

2. The method of claim **1**, wherein the height of the first edge of the porous polymer separator layer is at least 2% greater than the height of the second edge of the porous polymer separator layer.

3. The method of claim **1**, wherein the heights of each of the first and second edges of the porous polymer separator layer range from about 10 μm to about 50 μm.

4. The method of claim **1**, wherein step (b) comprises heating the polymer composition to form a melt of the polyolefin during formation of the film, and wherein step (c) comprises (c1) annealing the film and (c2) uniaxially stretching the film.

5. The method of claim **4**, wherein the polymer composition comprises about 20 wt. % to about 100 wt. % of the polyolefin.

6. The method of claim **4**, wherein the polymer composition further comprises particles having a higher melting temperature than the polyolefin, and wherein the particles become dispersed within the melt of the polyolefin during formation of the film.

7. The method of claim **1**, wherein the polymer composition comprises the polyolefin and a plasticizer, wherein step (b) comprises heating the polymer composition to form a mixture of a melt of the polyolefin and the plasticizer during formation of the film, and wherein step (c) comprises (c1) biaxially stretching the film and (c2) extracting the plasticizer from the film.

8. The method of claim **7**, wherein the polymer composition comprises about 20 wt. % to about 95 wt. % of the polyolefin, and about 5 wt. % to about 70 wt. % of the plasticizer based on a weight of the polyolefin and the plasticizer.

9. The method of claim **7**, wherein the polymer composition further comprises particles having a higher melting temperature than the polyolefin, and wherein the particles become dispersed within the mixture of the melt of the polyolefin and the plasticizer during formation of the film.

10. The method of claim **7**, wherein the plasticizer is an organic liquid that can be absorbed into the polyolefin and, once absorbed, lowers the glass transition temperature of the polyolefin.

11. The method of claim **10**, wherein the plasticizer is mineral oil.

12. The method of claim **10**, wherein the plasticizer is paraffin oil.

13. The method of claim **1** further comprising:

(d) disposing the porous polymer separator layer between a negative electrode and a positive electrode to assemble an electrochemical battery cell of a secondary liquid-electrolyte battery.

14. The method of claim **13**, wherein the secondary liquid-electrolyte battery is a lithium ion battery.

15. The method of claim **1**, wherein step (a) comprises:

extruding the film along an extrusion direction such that a width of the film oriented transverse to the extrusion direction is defined by the cross-section that includes the first and second edges of the film.

16. An electrochemical battery cell of a lithium ion battery that includes a negative electrode, a positive electrode, and a porous polymer separator layer disposed between the negative electrode and the positive electrode, the porous polymer separator layer functioning to electrically insulate the negative and positive electrodes yet communicate lithium ions between the electrodes when infiltrated with a liquid electrolyte, and wherein the porous polymer separator layer is formed according to the method of claim **1**.

17. A method of making and using a porous polymer separator layer that may function as all or part of a separator for use in an electrochemical battery cell of a lithium ion battery, the method comprising:

(a) extruding a film from a polymer composition that comprises a polyolefin, the film having a cross section including a first edge and second edge opposite the first edge, each of the first and second edges having a height that defines a thickness of the film at the first and second edges, respectively, and wherein the height of the first edge is greater than the height of the second edge such that the film has a thickness at the cross-section that decreases from the first edge to the second edge; and

(b) deriving a porous polymer separator layer from the film, the porous polymer separator layer having a cross-section that comprises a first edge derived from the first edge of the film, a second edge derived from the second edge of the film, the first and second edges of the porous polymer separator layer having a similar height relationship as the first and second edges of the film, and opposed first and second faces that extend between the first and second edges of the porous polymer separator layer, the porous polymer separator layer further having a porosity distributed throughout the separator layer such that, when the separator layer is infiltrated with a liquid electrolyte, lithium ions can be communicated through the thickness of the separator layer coextensively between the first and second opposed faces.

18. The method of claim **17**, wherein the height of the first edge of the porous polymer separator layer is at least 2% greater than the height of the second edge.

19. The method of claim **17**, wherein the porosity of the porous polymer separator is distributed evenly throughout the separator layer and ranges from about 30% to about 90%.

20. The method of claim **17**, further comprising:

(c) disposing the porous polymer separator layer between a negative electrode and a positive electrode to assemble an electrochemical battery cell of a lithium ion battery; and

(d) infiltrating the porous polymer separator layer with a liquid electrolyte.

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